

PROFESSIONAL PROFILE

Electrical and electronics engineer with over five years in electric powertrain, covering both sides of the supply chain. Four years developing traction machines at a motor manufacturer, rising from design engineer to R&D Lead, then a deliberate move to the OEM side for vehicle-level experience. Designed, validated and industrialised electric machines for e-mobility and industrial programmes including TOGG, ISUZU, ASELSAN-KARSAN and Nordex. Now responsible at Hyundai Motor Manufacturing Turkey for charging-efficiency validation, ECU and CAN diagnostics, and quality assurance on new IONIQ models.

CORE COMPETENCIES

Electric Machine Design	IPM, synchronous reluctance & induction machines · electromagnetic topology · FEA-based optimisation
E-Mobility Systems	Electric Drive Unit (EDU) integration · traction motor sizing · inverter & battery supplier coordination
Test & Validation	Prototype validation · dyno & performance testing · charging efficiency
Vehicle Diagnostics	CAN bus analysis · ECU fault diagnosis · in-vehicle data logging · ECU measurement & calibration
Quality & Compliance	Quality management systems · customer & certification audits · countermeasure processes
Leadership	R&D team building · project leadership · training & technical documentation

PROFESSIONAL EXPERIENCE

Quality Engineer — EV Performance & Diagnostics

2025 – Present

Hyundai Motor Manufacturing Turkey · Kocaeli, Türkiye

- Lead charging-efficiency and performance validation for the IONIQ series against global EV standards and internal quality gates.
- Trace vehicle-level electrical faults to root cause with Vector CANoe/CANalyzer, Intrepid Vehicle Spy and ETAS INCA, analysing recorded CAN traffic and ECU behaviour to close issues raised in production and validation.
- Quantify charging performance, efficiency losses and power quality through high-precision measurement with Hioki power analysers.
- Run quality assurance for the BJ new-model programme as project leader, aligning R&D and production teams from technical validation to launch readiness.
- Manage the Global Quality Management System (GQMS) improvement pipeline, turning shop-floor findings into countermeasures that lift production efficiency and product quality.

AEMOT Electric Motors & Generators

2021 – 2025

R&D Lead

2025

- Built and led a multidisciplinary R&D team spanning design, simulation, test and technical documentation, and set development priorities across the electric machine portfolio.
- Owned the full development cycle for motors and generators, from electromagnetic design and simulation through prototyping, validation testing and industrialisation.
- Converted a diesel forklift to electric drive: specified and developed the dedicated traction motor, then coordinated the inverter and battery suppliers through to a validated prototype.

AEMOT Electric Motors & Generators (continued)**Senior R&D Engineer**

2023 – 2025

- Led the electromagnetic design and optimisation of industrial and e-mobility machines, from concept to design freeze.
- Prepared the technical documentation for international certification and customer audits, and led the on-site reviews.
- Delivered motor design, testing and commissioning within electric drive unit (EDU) programmes for electric vehicles.
- Delivered training on electric machine design and test methodology to universities, industry partners and internal engineering teams.

R&D Engineer

2021 – 2023

- Designed the electromagnetic topology of synchronous reluctance, induction and IPM machines in Motor-CAD, Altair Flux and Ansys Maxwell.
- Wrote the product datasheets and test reports for the electric machine range.
- Supported production and testing on the shop floor, including coil winding, rotor-stator assembly and prototype validation.

KEY PROJECTS

TOGG EV Traction Motor	Lead electrical engineer
ASELSAN-KARSAN Motor Project	Lead electrical engineer
ISUZU L7-Class Traction Motor	Project leader
Pilotcar Motor Project	Project leader
Nordex Wind Turbine Pitch & Yaw Motor	Lead electrical engineer

TECHNICAL SKILLS

Electric Machine Design	Ansys Maxwell, Altair Flux, Motor-CAD, Simcenter SPEED
CAN Analysis & Diagnostics	Vector CANoe, Vector CANalyzer, ETAS INCA, Intrepid Vehicle Spy (neoVI FIRE)
CAD	SolidWorks, AutoCAD
Systems & Reporting	IFS ERP, Microsoft Office

EDUCATION

B.Sc. Electrical and Electronics Engineering — Aksaray University	2015 – 2019
English Preparatory Year — Niğde University	2014 – 2015

LANGUAGES

Turkish	Native
English	Professional working proficiency